

COURSE PLANNING (RPKPS) UNIVERSITAS MULTIMEDIA NUSANTARA

VALIDATION PAGE

Course Name : Software Engineering & Project Management X CIMB NIAGA (MATCHING CURRICULUM))

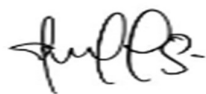
Course Code : IF 350

Course Coordinator : Eunike Endariahna S, S.Kom., M.T.I

Team of Lecturers :

NO	NO	NIK/NID	SIGN
1	Eunike Endariahna Surbakti, S.Kom., M.T.I.	066628	
2	Anak Agung Ngurah Ananda Kusuma, B.Eng., M.Eng., Ph.D.	L00044	
3	Dr. Sy. Yuliani Yakub, S.Kom., M.T	100008	

On behalf of the team,
Date: 25 July 2025



(Eunike Endariahna S)
Course Coordinator

Approved by
Date:



(Arya Wicaksana, S.Kom.,
M.Eng.Sc)
Head of Department/Program

Has been checked and considered comply
to UMN standard
Date:



(Dr. Maria Irminda Prasetyowati, S.Kom.,
M.T.)
Ex-Officio Informatika



COURSE PLANNING (RPKPS) UNIVERSITAS MULTIMEDIA NUSANTARA

SUBJECT : SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
 COURSE CODE / SKS : IF350/ 3 SKS
 SEMESTER : 3
 PREREQUISITE : -
 COURSE STATUS : MUST

A. COURSE DESCRIPTION

This course covers the principles of software engineering and project management, following industry best practices. Students learn to design, develop, test, and present software solutions using an Agile (Scrum) approach and implement automated testing. In collaboration with CIMB Niaga through the Matching Curriculum scheme, students are encouraged to demonstrate creativity, technological entrepreneurship, and technical skills in producing original digital products (IPR) and presenting their results to industry partners.

B. LEARNING OUTCOME

B.1. Graduate Learning Outcomes (CPL) Assigned to Courses

KKNI Level: 6

ELO 3	Professional Skills (Entrepreneurial spirit in technology domain) Able to demonstrate entrepreneurial spirit in the field of technology through innovation, creativity and communication skills
ELO 7	Software Design and Implementation Abilities Able to design, build, and maintain software solutions in accordance with software engineering standards and practices.

B.2. Course Learning Outcomes (CPMK)

After completing this course, science and technology and theory can be used in the engineering profession at the junior level with the following competencies:

ELO 3	CLO030	Explains the basic concepts and processes of software engineering and the role of project management in software development.
ELO 7	CLO050	Applying Agile methods, requirements analysis, design, and software testing collaboratively
ELO 7	CLO051	Produce innovative and relevant software solutions to user and industry needs and present them professionally

B.3. Sub-Course Learning Outcomes (Sub-CPMK)

CCLO030	SubCLO0301	Explain the meaning and scope of software engineering and the role of project management (C2)
CLO030	SubCLO0312	Using and differentiating software development process models (C3)
CLO050	SubCLO0711	Applying requirements analysis techniques and compiling initial software designs using business diagrams (C3,C4)
CLO050	SubCLO0712	Conduct testing using appropriate strategies (e.g. Selenium) and adapt to changing project requirements (C4)
CLO050	SubCLO0713	Work effectively in teams with an Agile approach and complete project deliverables on time (C5)
CLO051	SubCLO0721	Produce innovative software, compile final reports, and present them to industry (C6)

A. LEARNING ANALYSIS*-Figure is attached-***B. TOPICS**

1. Introduction of Software Engineering
2. Project Methodologies and Processes
3. Traditional or Agile
4. Business Case and Diagram Modeling.
5. Project Infrastructure.
6. DFD, Flowcharts and UML Diagrams.
7. Scope and The Work Breakdown Structure.
8. Schedule and Budgeting.
9. IT Project Quality Management.
10. Managing Project Quality and Automation Testing



11. Managing Organizational Change, Resistance, & Conflict.
12. Project Risk Management.
13. Project Implementation, Closure, and Evaluation.
14. Pitching Presentation

C. EVALUATION

1. Attending lectures punctually is mandatory. Students will be considered absent if coming over the specified time.
2. Attending 14 lectures is mandatory. Attending a minimum of 11 from 14 meetings is required to be able to take the final test.
3. Attending exam in week 7 and 14
4. Attending final project presentation in week 7 and 14
5. Final grade is determined by following components:
 - a. Midterm Test : 30% (MT)
 - b. Final Test : 30% (FT)
 - c. Final Project : 40% (FP)

ELO	CLO	Sub-CLO	Final Project	Midterm Test	Final Test
ELO 3	CCLO030	SubCLO0301 Explain the meaning and scope of software engineering and the role of project management (C2)		10%	
ELO 3	CLO030	SubCLO0312 Using and differentiating software development process models (C3)		10%	
ELO 7	CLO050	SubCLO0711 Applying requirements analysis techniques and compiling initial software designs using business diagrams (C3,C4)	10%	10%	
ELO 7	CLO050	SubCLO0712 Conduct testing using appropriate strategies (e.g. Selenium) and adapt to changing project requirements (C4)	10%		10%
ELO 7	CLO050	SubCLO0713 Work effectively in teams with an Agile approach and complete project deliverables on time (C5)	10%		10%
ELO 7	CLO051	SubCLO0721 Produce innovative software, compile final reports, and present them to industry (C6)	10%		10%

FINAL GRADING :

Score	Alphabetical Grade	Numerical Grade	Remarks
85 – 100	A	4	Excellent
80 – 84,99	A-	3,7	Good
75 – 79,99	B+	3,3	
70 – 74,99	B	3,0	
65 – 69,99	B-	2,7	Satisfactory
60 – 64,99	C+	2,3	
55 – 59,99	C	2,0	
45 – 54,99	D	1,0	Poor
0 – 44,99	E	0	Very Poor
	F	0	Academic Violation

D. REFERENCE AND RESOURCES**-Main-**

1. Pressman, R. S., & Maxim, B. R. 2020. *Software Engineering: A Practitioner's Approach*. 9th edition (International Edition). New York: McGraw-Hill Education.
2. Marchewka, J. T. 2015. *Information Technology Project Management: Providing Measurable Organizational Value*. 5th edition. Hoboken, New Jersey: John Wiley & Sons, Inc.
3. Sommerville, I. 2016. *Software Engineering*. 10th edition. Harlow, England: Pearson Higher Education.
4. **Surbakti E E, Purwandari B, Solicah, Kumoralita L. Analysis of software development method selection: a case of a private financial. ICBIM'19 : Proceddings of the 3rd International Conference on Business and Information Management.** <https://doi.org/10.1145/3361785.3361806>

E. WEEKLY LESSON PLAN

Week 2	MK Learning Outcomes (Sub CPMK)	Tree & Sub Tree Discussion	Learning Forms and Activities	Time	Assessment/ Assessment			Reference
					Form Assessment and Assessment Templates	Indicator	Weight	
1.	Explain the meaning and scope of software engineering and the role of project management (C2)	Subject: Introduction of Software Engineering Sub Topics: 1. Understanding software 2. Understanding software engineering 3. Application of software engineering	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% MT	Book 1/Chapter 1/page 3-11, Book 2/Chapter 1/page 17-42
2.	Explain the meaning and scope of software engineering in project management (C2)	<u>Subject:</u> Project Infrastructure <u>Sub Topics:</u> Process Model	get Learning: Asynchronous Activity: Videos, quizzes, pre activities, tasks	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% MT	Buku1/Bab2/hal15-27, Buku2/Bab2/hal43-71
3.		<u>Subject:</u> Traditional or Agile <u>Sub Topics:</u>	Form of Learning: Face-to-Face Lectures Activity:	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas)	Assessment: (Formatif) Final project group	Accuracy in answering each question	1% MT	Book 1/Chapter 3/pages

	Using and differentiating software development process models (C3)	Types of traditional and agile development approaches	Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	bebas) 3x60' (tugas mandiri/metod enya bebas)	study Midterm writing test Grading system: Point E.			30-38, Book 2/Chapter 4/pages 101-137, Book 3/Chapter 4/pages 89-119
4.	Applying requirements analysis techniques and compiling initial software designs using business diagrams (C3, C4)	<u>Subject:</u> Business Case and Diagram Modeling. <u>Sub-Discussion Tree:</u> 1. Process Models 2. Design Diagrams 3. RE diagrams	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% MT	Book 1/Chapter 4/pages 40-65, Book 2/Chapter 5/pages 138-167, Book 3/Chapter 5/pages 120-146, Book 4/Chapter 7, pages 187-216
5.	Work effectively in teams with an Agile approach and complete project deliverables on time (C6)	<u>Subject:</u> Scrum Method and Collaboraction Team <u>Sub-Discussion Tree:</u> 1. Sprint 2. Daily Stand Uo 3. Tools : Jira/Trello	Form of Learning: Face-to-Face or Hybrid Lectures Activity: Face to Face: Case Study Presentation from CIMB Niaga Discussion Material: Questions and Answers	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% MT	Buku1/Bab9/hal167-183, Buku2/Bab5/hal138-167, Buku4/Bab10/Hal261-300

			Task: Attached					
6.	Applying requirements analysis techniques and compiling initial software designs using business diagrams (C3,C4)	<u>Subject:</u> Schedule and Budgeting <u>Sub-Discussion Tree:</u> 1. Gantt Chart 2. EVM 3. Story Points	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/melompok) 3x60' (tugas mandiri/metode bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% FP	Buku1/Bab9/hal184-201, Buku2/Bab5/hal138-167, Buku4/Bab10/Hal261-300
7.	Work effectively in teams with an Agile approach and complete project deliverables on time (C6)	<u>Subject:</u> Risk and People Management <u>Sub-Discussion Tree:</u> 1. Risks in software engineering 2. Risk identification 1. Mapping, monitoring, and risk management Leadership 2. Managing team in culture	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/melompok) 3x60' (tugas mandiri/metode bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% FP	Buku1/Bab9/hal184-201, Buku2/Bab5/hal138-167, Buku4/Bab10/Hal261-300

Mid-Semester Test							25%	W1-W7
Test model and form: Essay								
8.	Work effectively in teams with an Agile approach and complete project deliverables on time (C6)	<u>Subject:</u> IT Project Quality Management. <u>Language Sub-Tree:</u> 1. PMBOK 2. Continuous Development 3. Unit and Integration Testing	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% FP	Buku1/Bab22/hal46-494, Buku2/Bab8/hal226-249
9.	Conduct testing using appropriate strategies (e.g. Selenium) and adapt to changing project requirements (C4)	<u>Subject:</u> Software Testing <u>Language Sub-Tree:</u> 1. UAT 2. Automation Testing 3. Selenium and Katalon	Form of Learning: Face-to-Face and Hybrid Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1%FP	Book 1/Chapter 11/page 203-222, Book 2/Chapter 7/page 196-219
10.		<u>Subject:</u> Managing Organizational Change, Resistance, & Conflict. <u>Language Sub-Tree:</u>	Form of Learning: Asynchronous Activity: Videos, quizzes, pre activities, tasks	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test	Accuracy in answering each question	1% FP	Buku1/Bab31/hal68-701, Buku2/Bab22/hal64-656

		Change Management Plan			Grading system: Point E.			
11.		<u>Subject:</u> Care and Maintenance <u>Language Sub-Tree:</u> 1. Bug Fix 2. Proposal Documentation 3. Support	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Attached	3x50' (tatap muka) 3x60' (tugas terstruktur/mendoneya bebas) 3x60' (tugas mandiri/metode nya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% FP	Buku1/Bab22/hal 412-429, Buku19/Bab10/hal 285-299
12.	Produce innovative software, compile final reports, and present them to industry (C6)	<u>Subject:</u> Project Implementation and Closure <u>Language Sub-Tree:</u> HKI dan Project Implementation, Closure, and Evaluation	Form of Learning: Face-to-Face Lectures Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task:	3x50' (tatap muka) 3x60' (tugas terstruktur/mendoneya bebas) 3x60' (tugas mandiri/metode nya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Accuracy in answering each question	1% FP	Buku1/Bab37/hal81 8-837, Buku2/Bab20/hal58 0-599
13.		<u>Subject:</u> Evaluation Project	Form of Learning: Face-to-Face Lectures	3x50' (tatap muka) 3x60' (tugas	Assessment: (Formatif)	Group presentation	20% FP	Book1/Bab35/hal77 7-793, Book2/Ba

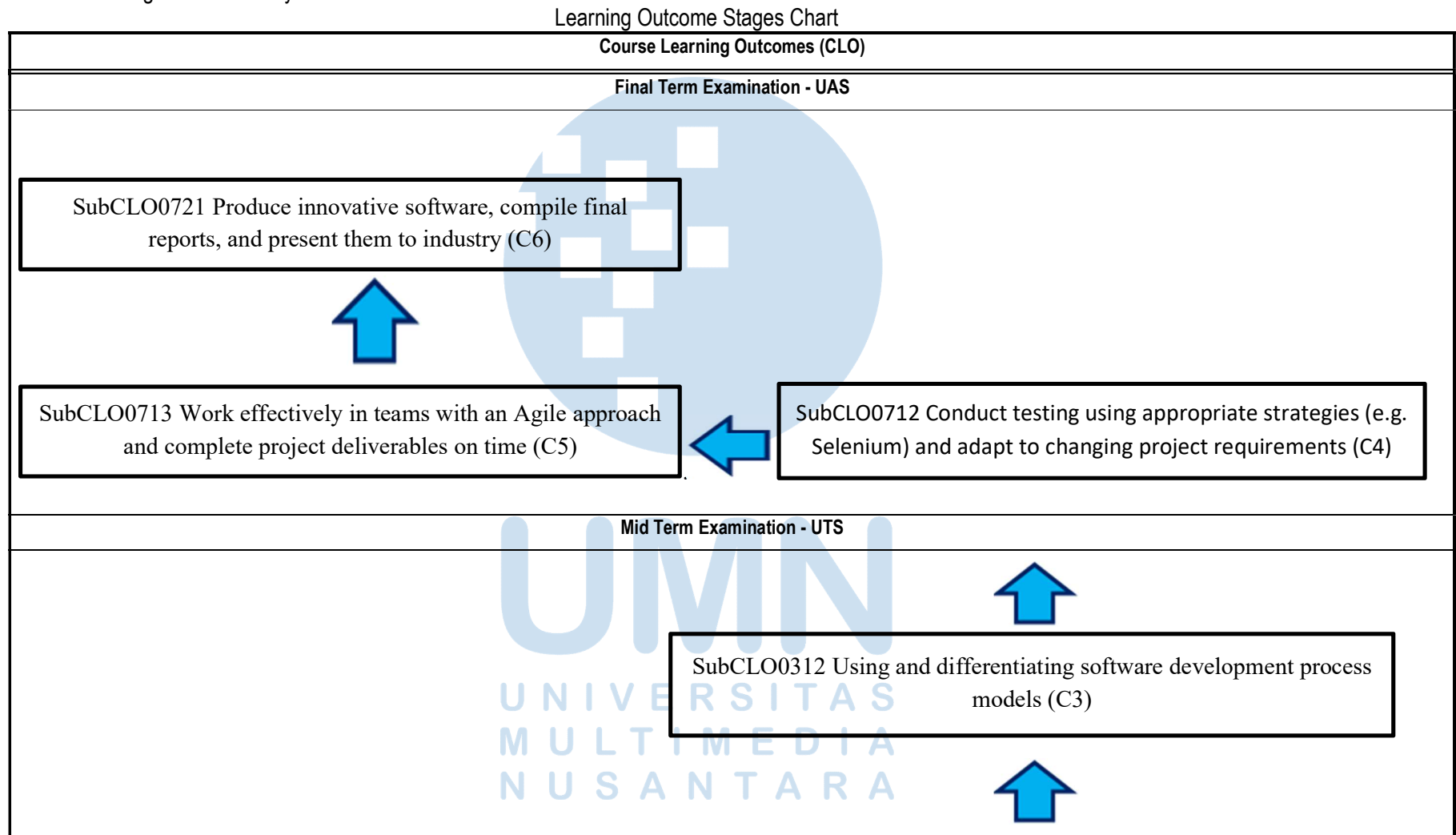
	Produce innovative software, compile final reports, and present them to industry (C6)	<u>Language Sub-Tree:</u> 1. Evaluation Project 2. Pitching Preparation	Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Project Performance Evaluation & Final Project Report Presentation	terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Final project group study Midterm writing test Grading system: Point E.			b24/hal70 0-716
14.	Produce innovative software, compile final reports, and present them to industry (C6)	<u>Subject:</u> Pitching Final Project <u>Language Sub-Tree:</u> Project Implementation, Closure, and Evaluation.	Form of Learning: Guest Lecture for all classes Activity: Face to Face: Presentation of Discussion Material: Questions and Answers Task: Presentation of projects to 3 Naga	3x50' (tatap muka) 3x60' (tugas terstruktur/metodenya bebas) 3x60' (tugas mandiri/metodenya bebas)	Assessment: (Formatif) Final project group study Midterm writing test Grading system: Point E.	Group presentation	5% FP	Buku1/Bab38/hal840-858, Buku2/Bab25/hal730-750
Final exams Exam model and format: Final Project Proposal							30%	W8-W14

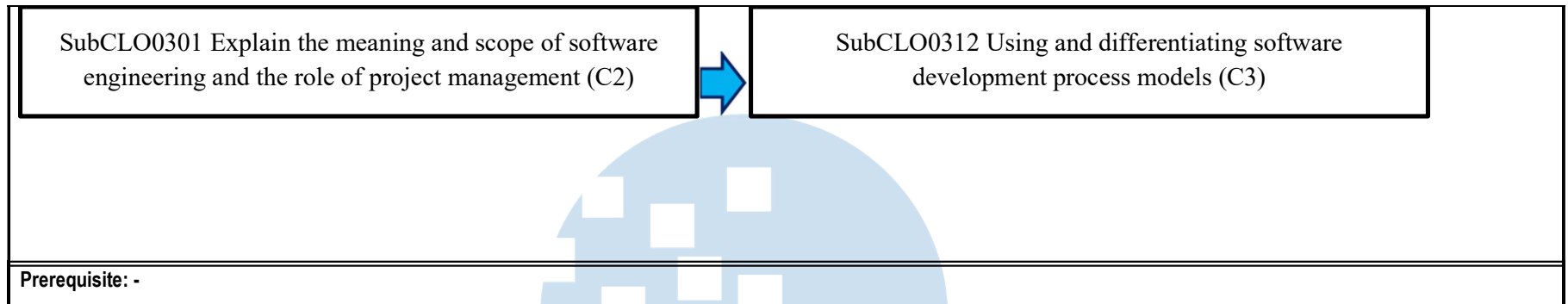
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F. Task/Project Details:**1. Task/Project no: 1 Week 4**

Subject	: Software Engineering and Project Management	Home MK	: IF350
Assignment Title	: Software Engineering Project Paper	Assignment Weight	: 40%
Sub-CPMK related tasks	: SUB CPMK 0721		
A. INDEPENDENT ACTIVITIES			
Description	Students are assigned to conduct research, study, and write papers related to projects that have been built and then register them with HKI.		
B. STRUCTURED ASSIGNMENT			
Task Form	: Software Engineering Project Paper		
Description	: Students are assigned to search, study, and create papers with the following directions: 1. Conduct a search for journals that are the target of the project paper that will be created. 2. Follow the directions and journal template provided by the editor of the target journal. 3. Create a paper that follows good and correct writing rules and is free from elements of plagiarism. 4. Papers are collected during the Final Semester Exam period.		
External Shape and Format	: <u>External Form:</u> Project Papers and Prototypes <u>External Format:</u> Outside of Copyright		
Indicators, Criteria, and Assessment Weights	: Indicator (Weight 40%) - Compliance with Report template(20%) - Use of standard and easy-to-understand language (10%) - Logical and systematic writing organization (5%) - Completeness of IPR documents (5%) - Up-to-date and appropriate references 10%)		
Terms and conditions regarding processing time	: Total processing time: with details: a. Literature study b. Preparation and creation of papers	3 weeks 1 week from the assignment date 2 weeks from the assignment date	
Others	Papers suspected of plagiarism will be subject to sanctions in accordance with the regulations in force at UMN.		
Reference	-		

Attachment: Learning Outcome Analysis





G. Revision History

Kode MK	Revisi ke	Tanggal Berlaku	Perubahan
IF 350	1	2022	Correction course name Software Engineering andproject Management
If 350	2	2024	Add final project
IF 350	3	2025	Adapt new curriculum with cimb niaga and translate Kurikulum Matching